



Energy Precisions

Engineered hybrid solar · LiFePO₄ lithium storage · Full installation

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Our only office. Site surveys and installs across Ghana are booked from Accra.

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HYBRID LITHIUM SOLAR PACKAGES

WITH LiFePO₄ LITHIUM STORAGE

Hybrid solar with lithium backup (grid + generator ready). Turnkey supply, mounting, protection, commissioning & monitoring for Ghana homes and businesses.

Turnkey hybrid packages — six tiers

How to read these packages

- KVA on the badge = your load tier (how much you plan to run at once), not the inverter brand size.
- Watts shown ($\sim 0.85 \times \text{kVA}$) = continuous planning ceiling — stagger AC, iron, and heaters; do not run every heavy load together.
- Inverter line = equipment we stock; it may be larger than the kVA badge for starting surges and reliability.
- Panel count = sized to the kVA load tier, not to fill a larger inverter — survey may add or reduce modules.
- Hybrid = solar + lithium + grid (and generator where fitted); backup hours depend on battery size and your night load.
- Turnkey price is from our Accra office; final BOM and price are confirmed after a free site survey.

6.5 KVA SYSTEM

Planned load $\sim 5,500$ W · stagger heavy items

ESSENTIAL

FROM

GHS 140,900

Entry tier · flats & small homes

- ✓ 6.5 kW hybrid inverter (1)
- ✓ 16 kWh LiFePO₄ lithium battery (1)
- ✓ Battery management / monitoring (1)
- ✓ 570W tier-1 solar panels (15)
- ✓ DC protection, changeover & AC distribution
- ✓ Mounting structure (roof)
- ✓ Cables, MC4, earthing & commissioning

6.5 KVA tier · matching 6.5 kW inverter · 15 panels · no AC in this tier.

TYPICAL LOADS (STAGGER HEAVY ITEMS)

Fridge, TV, standing fans, router, 15–20 LED lights, phone and laptop charging, blender or kettle (one at a time). No air conditioning — add a higher tier for AC.

8 KVA SYSTEM

+25% inverter headroom · AC motor starts
Planned load $\sim 6,800$ W · stagger heavy items

HOME

FROM

GHS 194,900

Popular family home

- ✓ 10 kW hybrid inverter (1)
- ✓ 16 kWh LiFePO₄ lithium batteries (2)
- ✓ Battery management / monitoring (1)
- ✓ 570W tier-1 solar panels (19)
- ✓ DC protection, changeover & AC distribution
- ✓ Mounting structure (roof)
- ✓ Cables, MC4, earthing & commissioning

8 KVA ($\sim 6,800$ W). 10 kW inverter for AC starts · 19 panels sized for 8 KVA load.

TYPICAL LOADS (STAGGER HEAVY ITEMS)

Fridge, rice cooker, blender, iron (staggered), 1 split AC (1–1.5 hp), 25 LED bulbs, 4–5 fans, 2 TVs, home Wi-Fi, 2–3 laptops or computers, printer.

Packages — Plus to Power

10 KVA SYSTEM

Planned load ~8,500 W · stagger heavy items

PLUS

FROM

GHS 229,900

Large home or small office

✓ 10 kW hybrid inverter (1)

✓ 16 kWh LiFePO₄ lithium batteries (3)

✓ Battery management / monitoring (1)

✓ 570W tier-1 solar panels (23)

✓ DC protection, changeover & AC distribution

✓ Mounting structure (roof)

✓ Cables, MC4, earthing & commissioning

10 KVA / 10 kW aligned · ~8,500 W continuous · stagger two AC units.

TYPICAL LOADS (STAGGER HEAVY ITEMS)

2 fridges or fridge + freezer, rice cooker or microwave, blender, iron (staggered), up to 2 split AC (1-1.5 hp, not together on full cool), 30 LED bulbs, 4-5 fans, 2 TVs, small office (PCs, printer, router).

12 KVA SYSTEM

+8% inverter headroom · dual 6.5 kW sync

PRO

FROM

GHS 280,900

Executive home · boutique office

✓ 6.5 kW hybrid inverters (2, synchronized)

✓ 16 kWh LiFePO₄ lithium batteries (4)

✓ Battery management / monitoring (1)

✓ 570W Jinko / Longi solar panels (28)

✓ DC protection, dual MPPT where required, changeover

✓ Mounting structure (roof)

✓ Cables, MC4, earthing & commissioning

12 KVA (~10,200 W) · dual 6.5 kW (~13 kW) · 28 panels for 12 KVA load.

TYPICAL LOADS (STAGGER HEAVY ITEMS)

2 fridges/freezers, kitchen blender, rice cooker, iron and heater (staggered), 2-3 split AC (1-1.5 hp — alternate with kitchen peaks, not all on full cool), 30+ LED bulbs, 4-5 fans, 2-3 TVs, home office or shop POS and computers.

15 KVA SYSTEM

+33% inverter headroom · dual 10 kW sync

COMMERCIAL

FROM

GHS 353,900

Guest house · shop · church hall

✓ 10 kW hybrid inverters (2, synchronized)

✓ 16 kWh LiFePO₄ lithium batteries (5)

✓ Battery management / monitoring (1)

✓ 570W tier-1 solar panels (35)

✓ AC/DC distribution boards & changeover

✓ Mounting structure (roof)

✓ Cables, MC4, earthing & commissioning

15 KVA (~12,800 W) · dual 10 kW sync · 35 panels (less than 20 KVA tier).

TYPICAL LOADS (STAGGER HEAVY ITEMS)

Several guest rooms, commercial fridge, kitchen appliances (staggered), up to 4 split AC (1-1.5 hp) with load management, sound system, reception PCs, security and corridor lighting, small water pump or booster (if within survey load).

20 KVA SYSTEM

POWER

FROM

GHS 447,900

Hotel wing · office block · multi-tenant

✓ 10 kW hybrid inverters (2, synchronized)

✓ 16 kWh LiFePO₄ lithium batteries (6)

✓ Battery management / monitoring (1)

✓ 570W tier-1 solar panels (46)

✓ AC/DC distribution boards & changeover

✓ Mounting structure (roof)

✓ Cables, MC4, earthing & commissioning

20 KVA (~17,000 W) · dual 10 kW · 46 panels · load schedule on survey.

TYPICAL LOADS (STAGGER HEAVY ITEMS)

Multiple guest-room AC (1-1.5 hp, staggered across rooms), commercial kitchen peaks, laundry alternated with AC, fridges, IT/server room on dedicated circuit, event or hall lighting — engineered load schedule on survey.

Why Energy Precisions?

Turnkey engineered packages — not guesswork. Every installation includes commissioning, monitoring setup where specified, and electrical protection suited to Ghana site conditions.

WHAT EVERY PACKAGE INCLUDES

- Hybrid inverter (grid + battery backup topology)
- LiFePO₄ lithium storage — daily cycling, longer service life
- Tier-1 PV modules (Jinko / Longi class)
- Roof mounting, DC / AC cabling, earthing & hazard labels
- Changeover / protection — handover-ready boards
- Remote monitoring & battery management as specified per tier
- Installation testing & client handover checklist

BEFORE YOU ORDER

- Hybrid systems combine solar, lithium storage, and ECG/grid (generator-ready where fitted). Full off-grid only when engineered on survey.
- Connected load ceilings use ~0.85 power factor. Stagger split AC, iron, kettle, and water heater — never assume all peaks run together.
- A larger inverter than the kVA badge is normal: it covers motor starts and growth while you still plan within the stated watt ceiling.
- Solar panel counts follow the package load tier (kVA), not the inverter nameplate — avoids paying for PV you cannot use on that load.
- Storage uses stocked 16 kWh LiFePO₄ modules; night backup hours depend on your load — survey confirms module count.
- Commercial and Power tiers need a load schedule on survey; brochure lists are typical examples, not unlimited simultaneous use.
- Larger projects receive engineered BOM, load analysis, and itemised quotation from Energy Precisions.
- Typical payment: 30% deposit · 40% on delivery · 30% on commissioning (negotiable for commercial clients).

THE ENERGY PRECISIONS DIFFERENCE

Site survey first

We confirm roof area, cable routes, and your real load before final BOM.

Transparent engineering

Load analysis, string planning, and itemised quotations.

Turnkey pricing

One price for supply and professional install.

Accra office, Ghana-wide

Nationwide surveys and installs by appointment.

Warranty & service: Premium 16 kWh LiFePO₄ batteries · 5-year battery warranty (manufacturer) · 2-year workmanship on installation.

Pricing validity: Prices valid until December 2026. Final design confirmed after site survey.

Book a free site assessment

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